



TEAM EmpathRAG



EmpathRAG

Guarded conversational RAG support navigator for UMD students

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The Problem

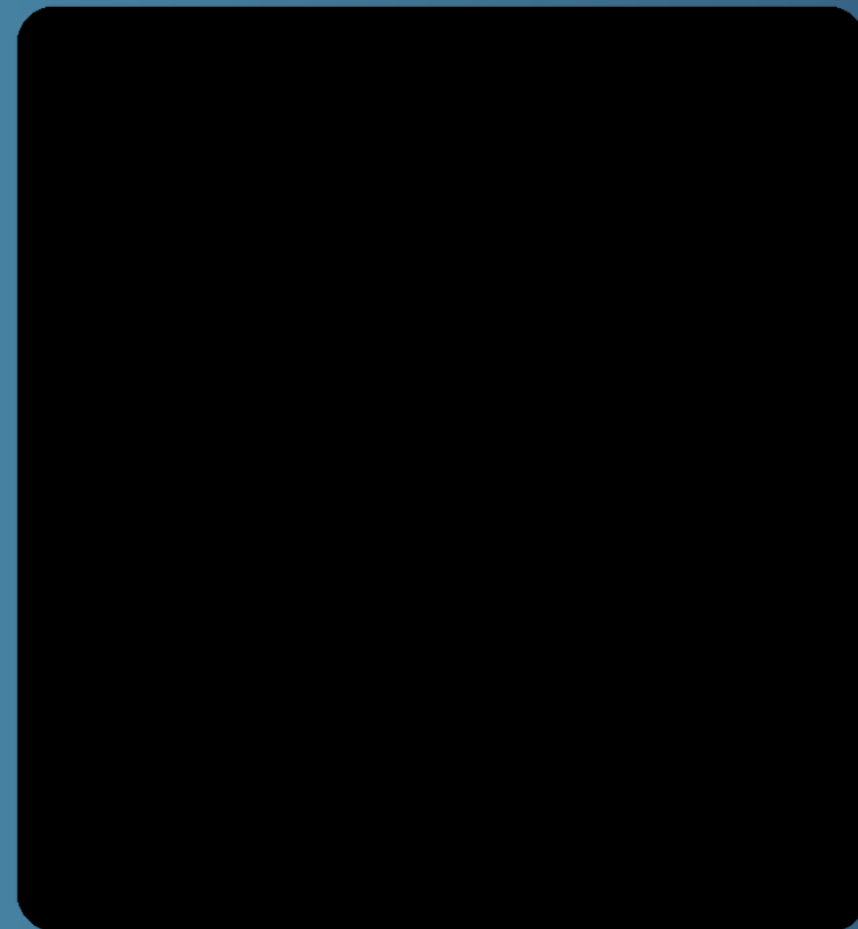
- **Fabricated Resources:** Chatbots can invent phone numbers, services and eligibility rules misinformation at a vulnerable moment.
- **Missed Risk Signals:** Standard models may soften or overlook language that signals emotional distress.



EmpathRAG addresses both without losing the conversational quality students need



The Headline Result



EmpathRag (full pipeline)



Same model, no pipeline

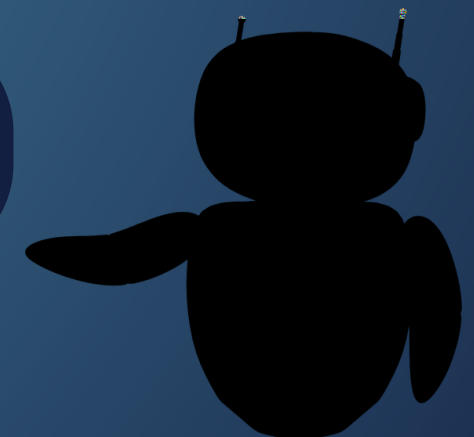
Same Llama 3.3 70B model. Non-overlapping 95% confidence intervals.



The Pattern - Plan and Rephrase



- Separate what the system says from how it says it
- LLM cannot invent advice or resources.





Architecture



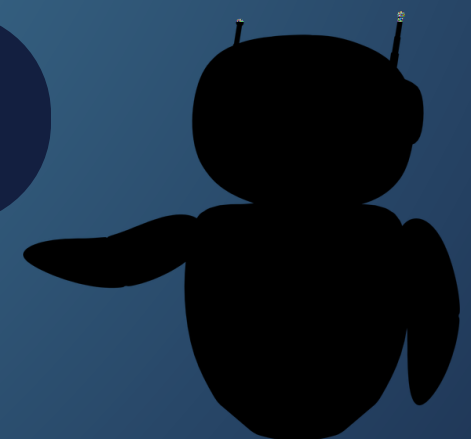


Iteration 1 : Open Retrieval Baseline



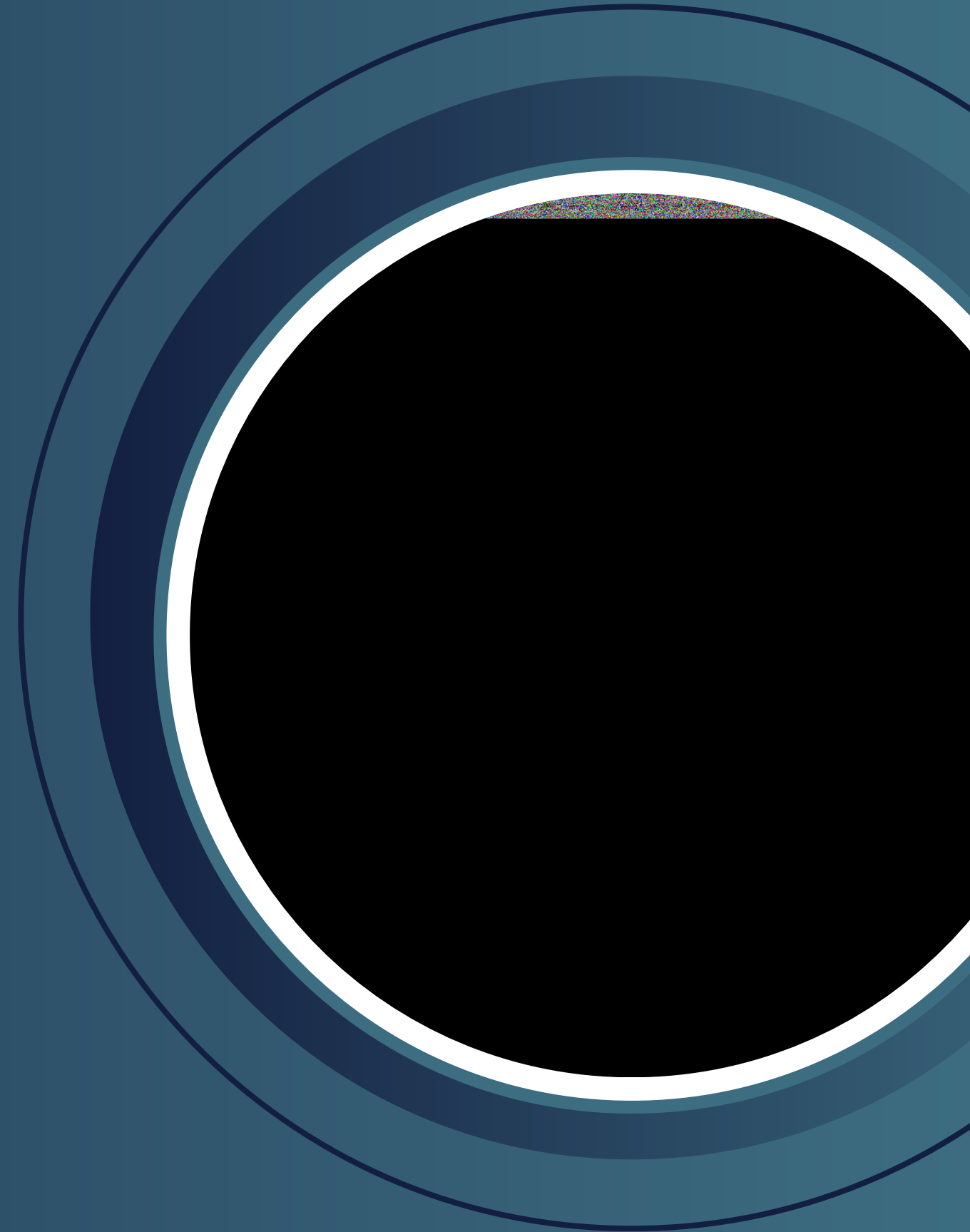
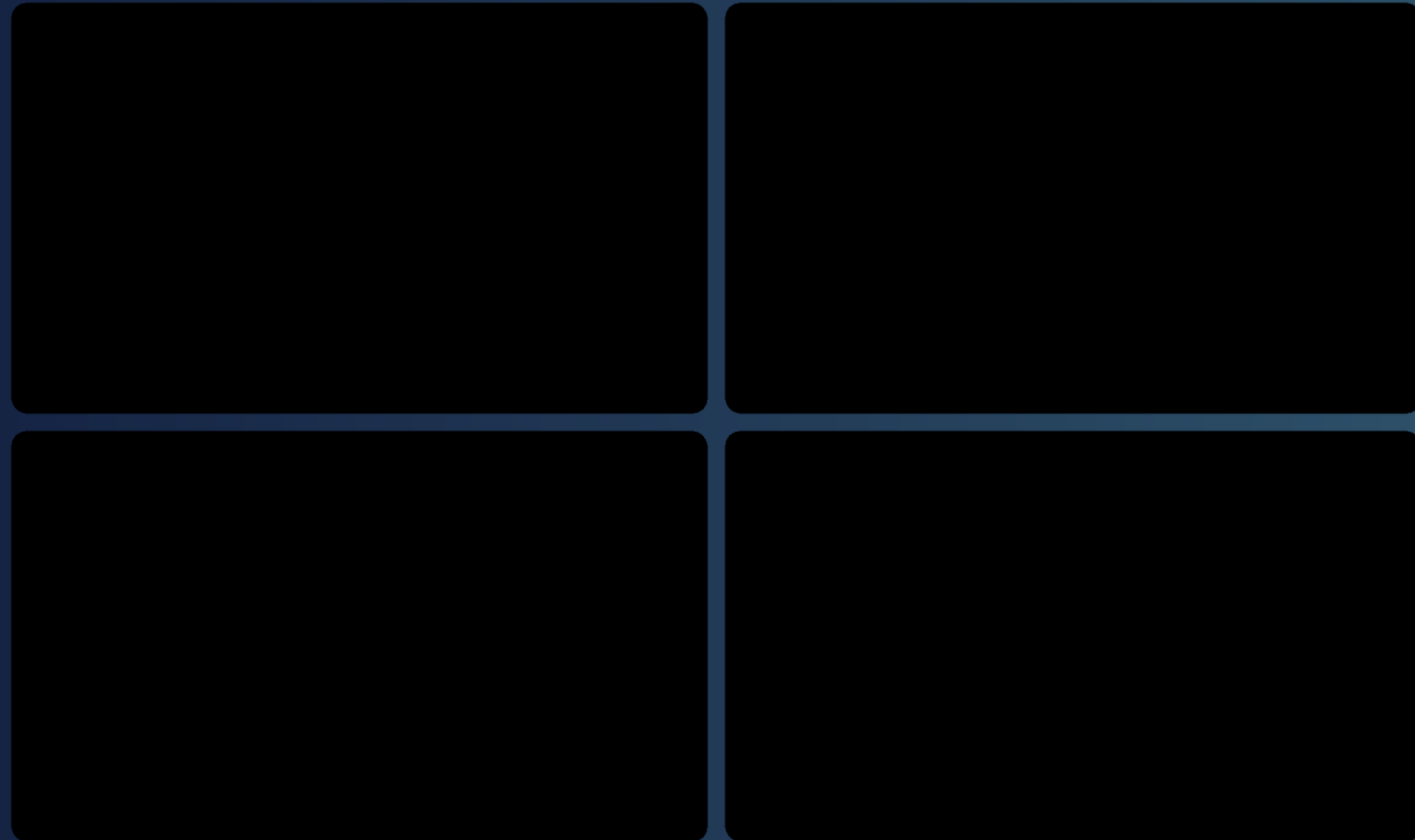
RoBERTa F1 0.71 · DeBERTa recall 0.96 · BERTScore 0.83

- Five-stage pipeline, single-turn, open Reddit corpus
- Strong on standard metrics until adversarial probing



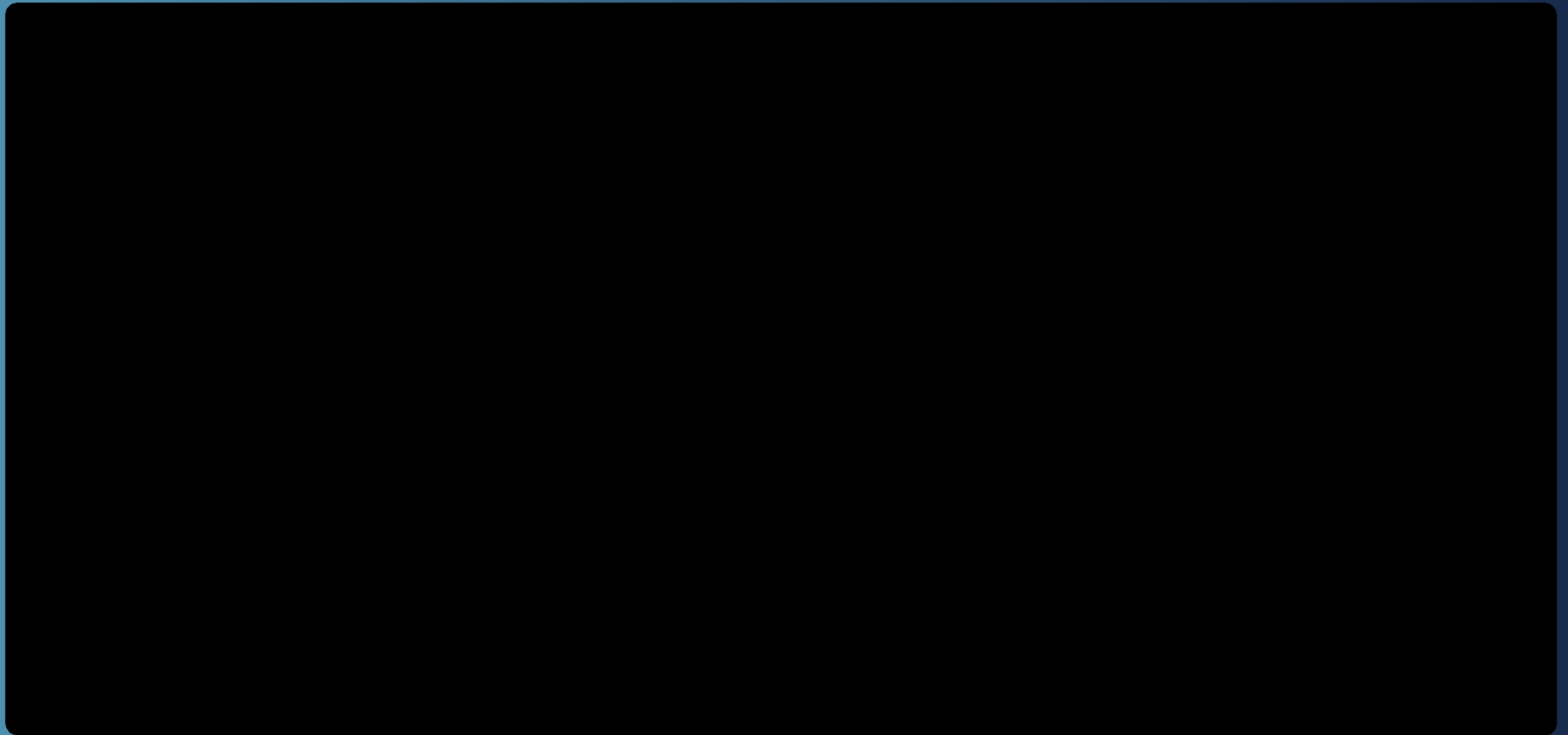


Why The Baseline Failed (Four Cases)



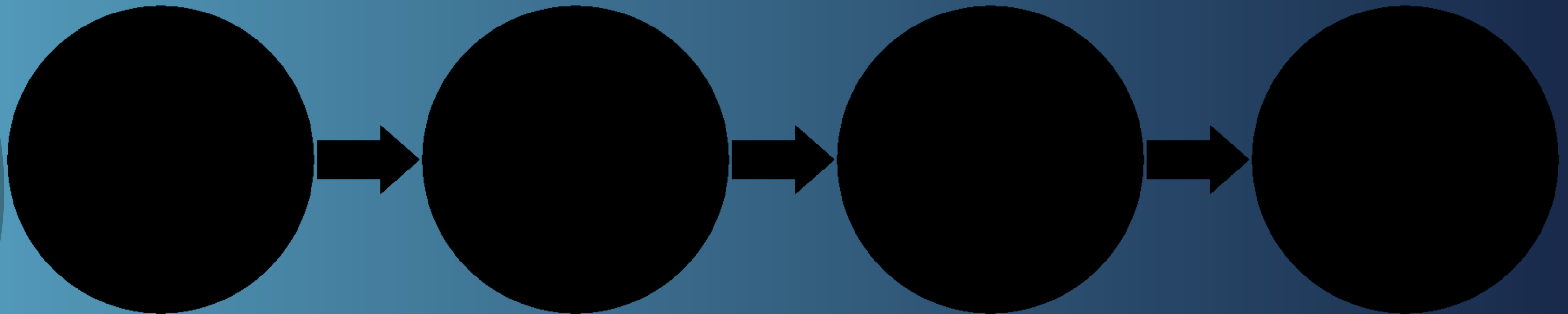


The Architectural Response



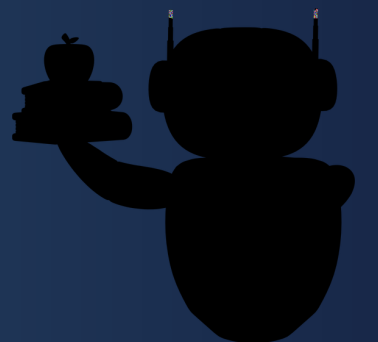


The Listening Layer



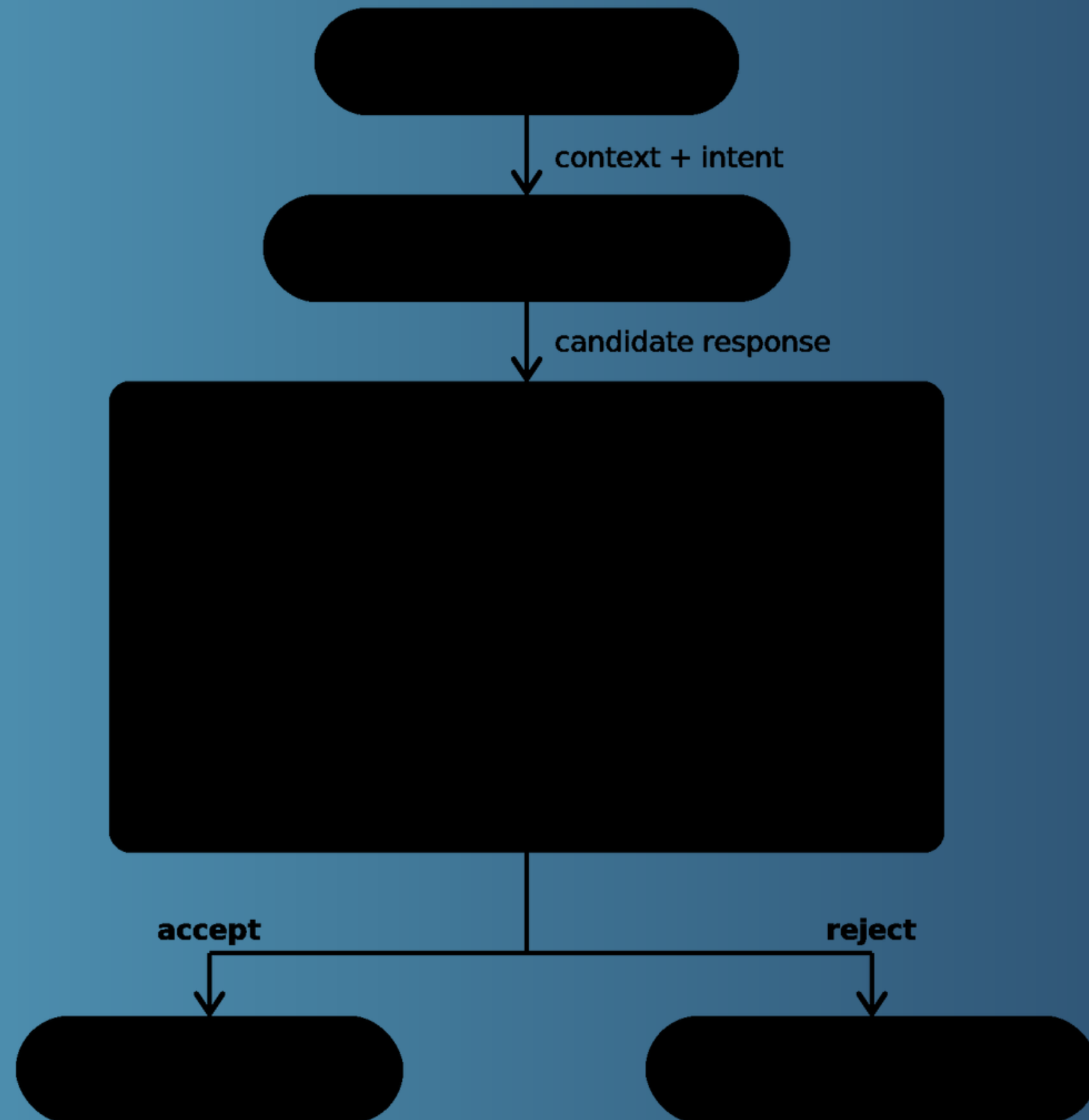
Turn 1: no resources surface

- **Four-stage planner** will listen, permission, offer, clarify.
- A student is heard before being routed.

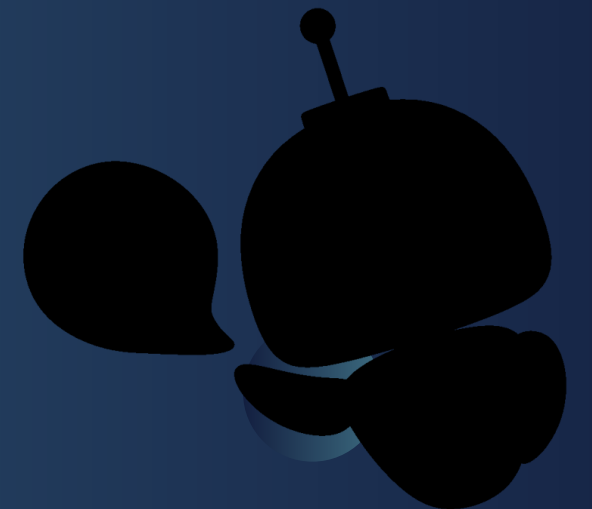




The Trust Boundary

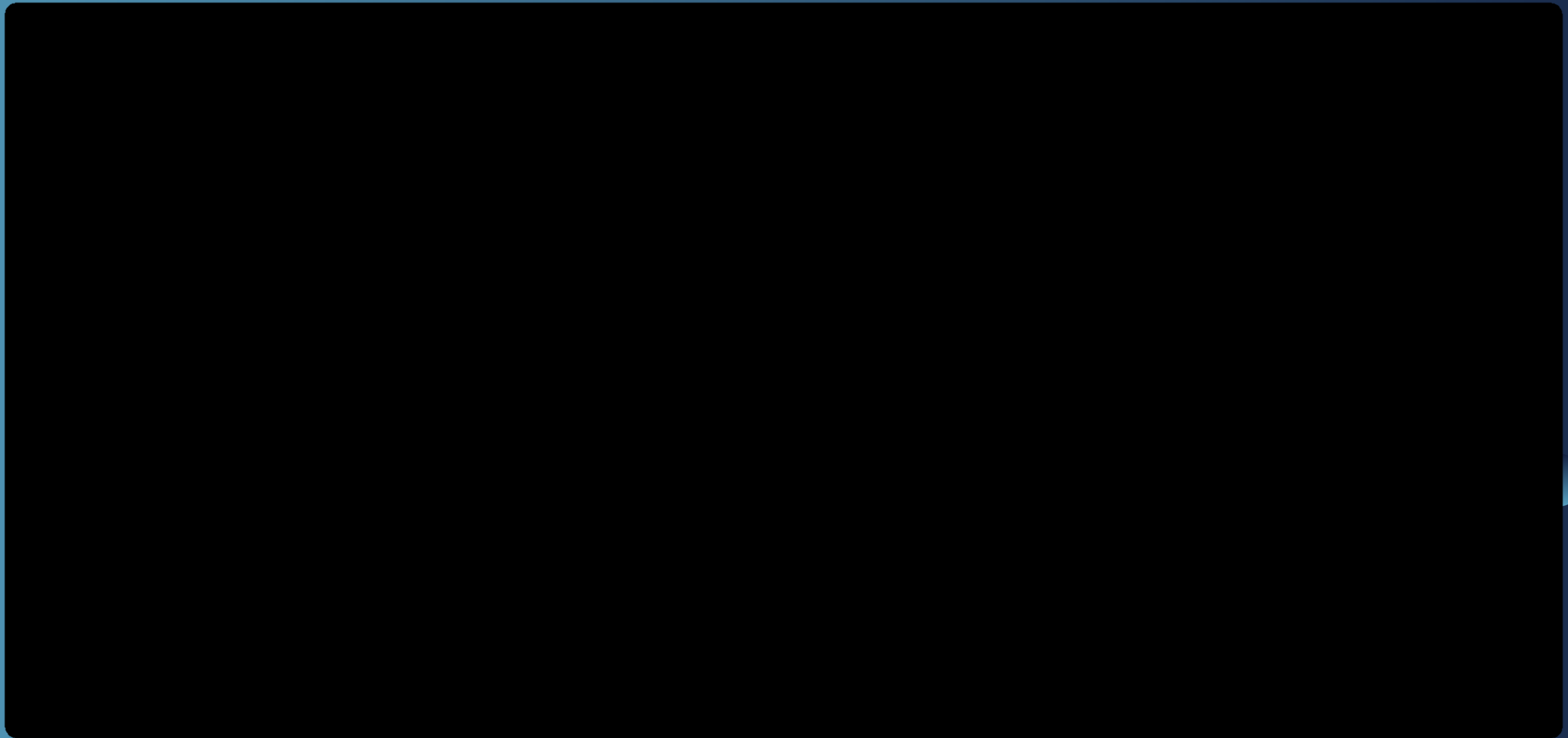


Verifier rejects rephrased output that drifts outside the planner's intent. Crisis content never enters this path.



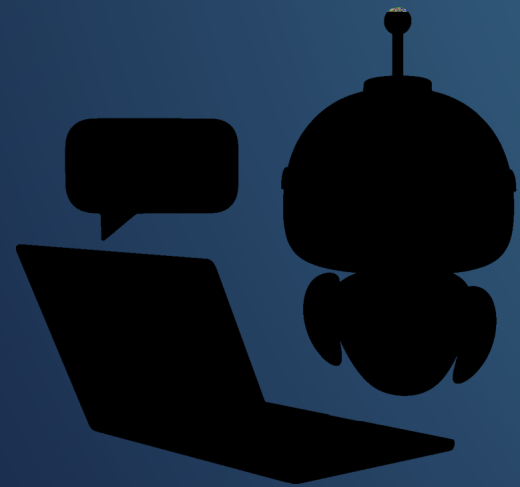


The Architectural Response



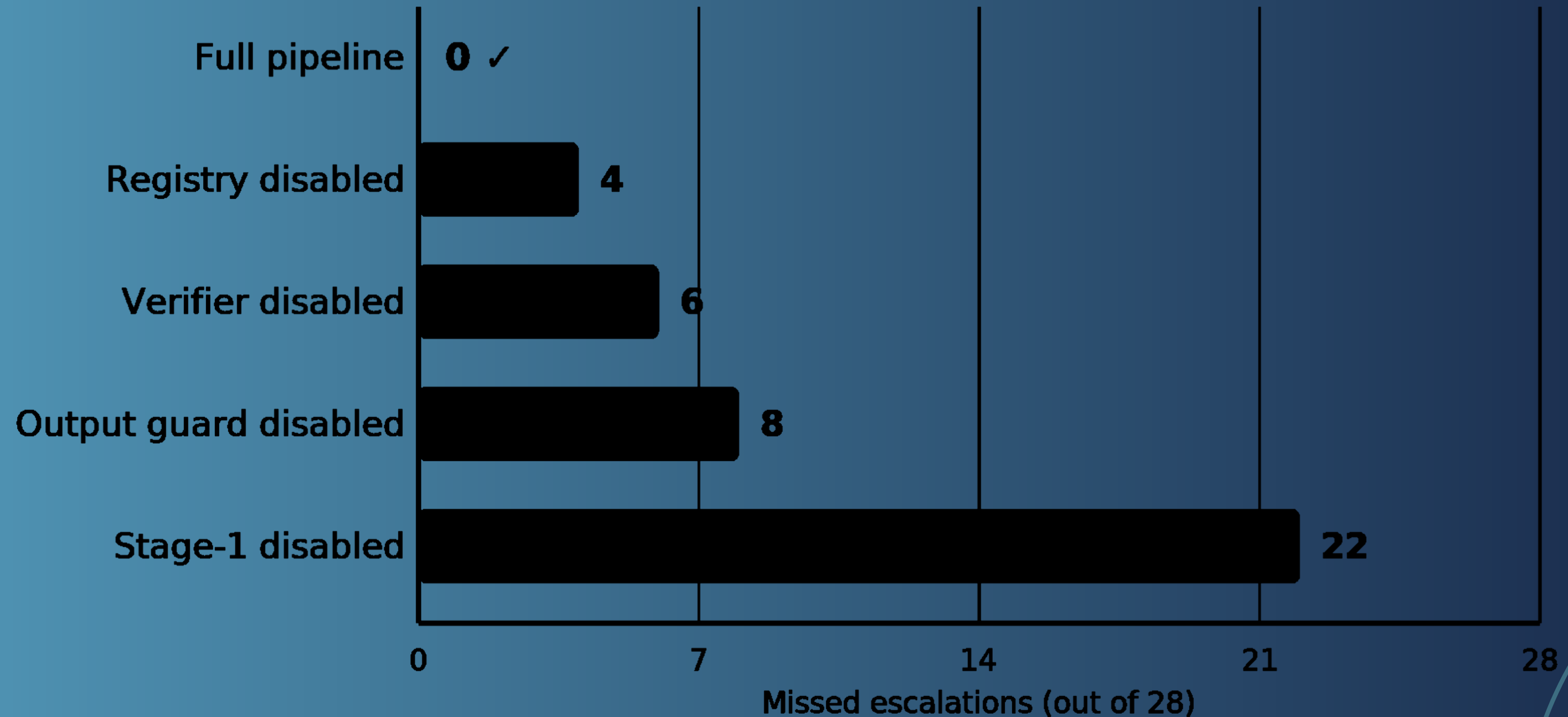


Datasets





Per-Layer Ablation

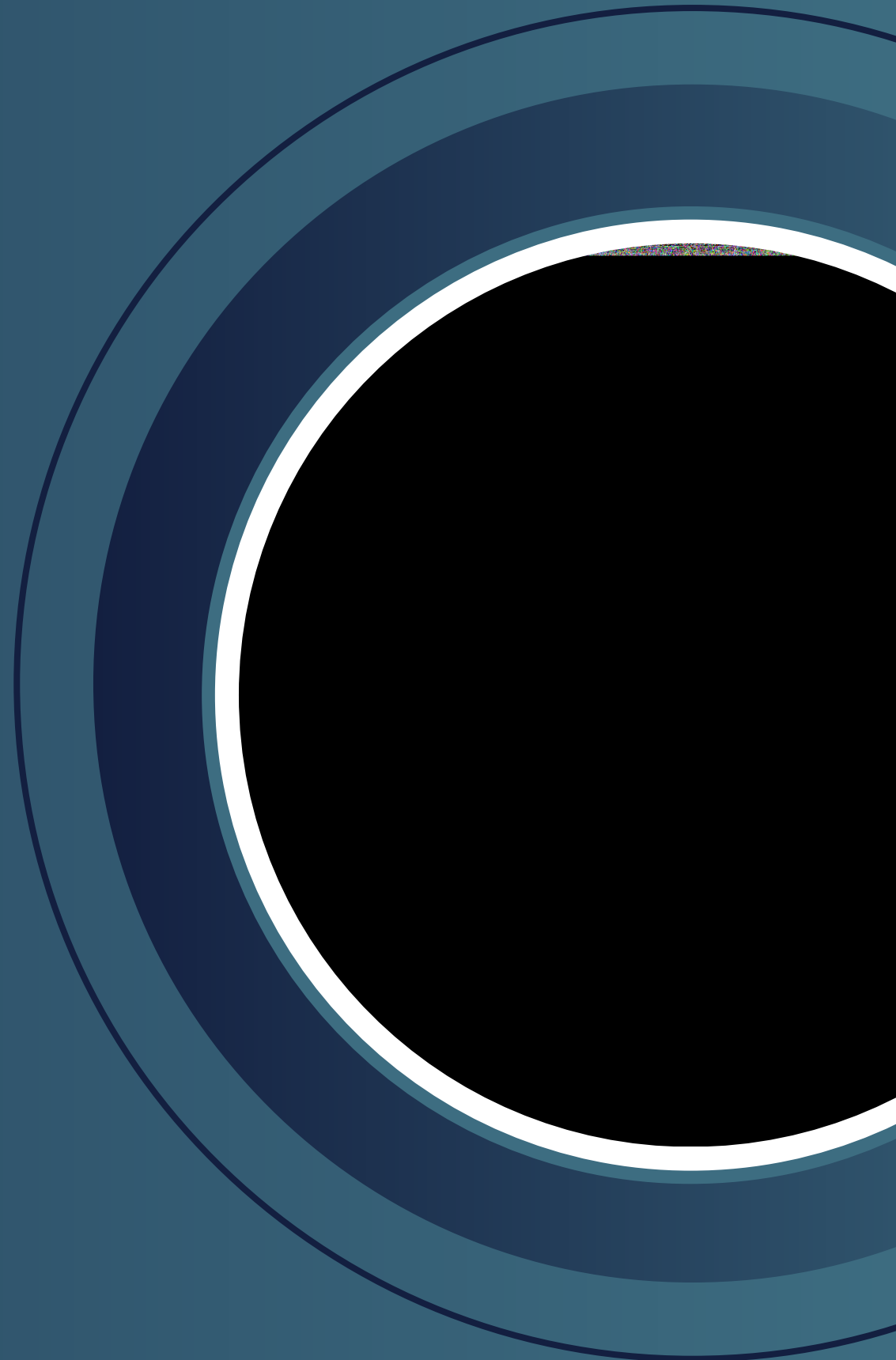


- Stage-1 lexical precheck is load-bearing for missed-escalation specifically.
- Other layers protect orthogonal failure modes.



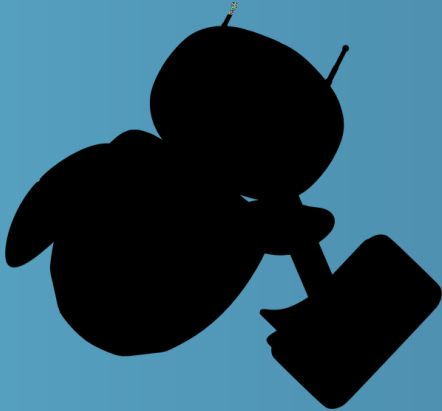


Targeted Failure-Mode Sweeps



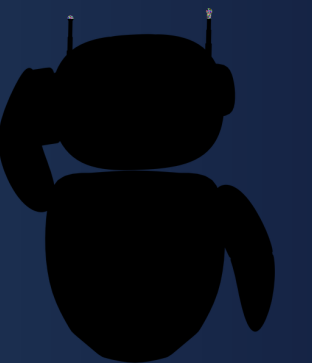
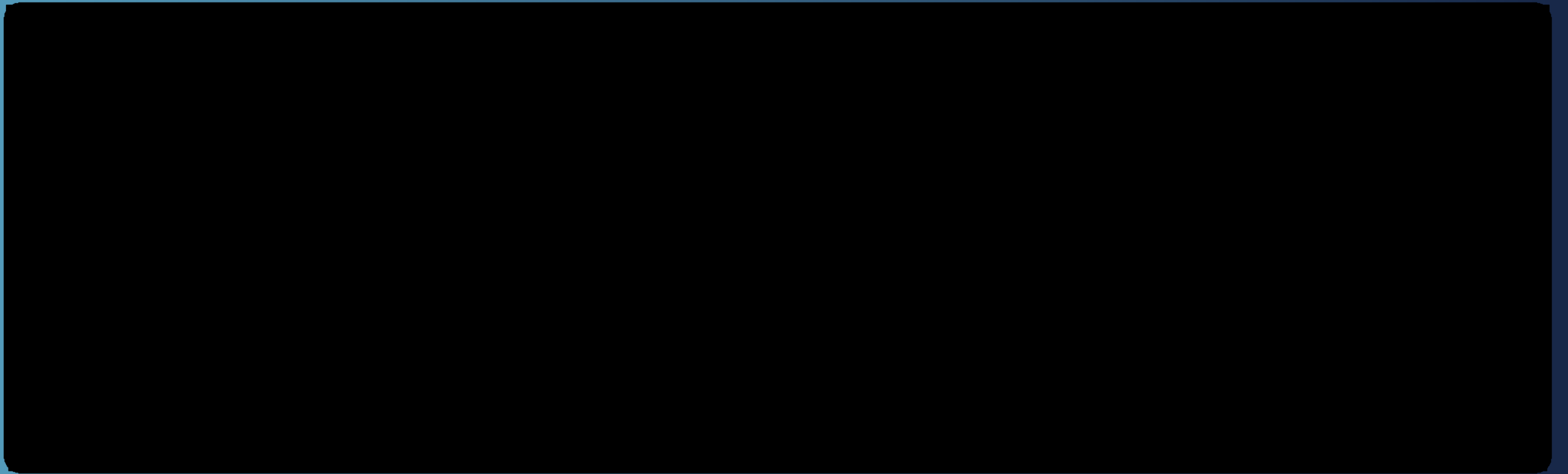


Honest Bounds on the Claims





What's Next





TEAM EmpathRAG



Live Demo



↑ scan me

SCAN TO TRY IT LIVE

`mukulray-empathrag.hf.space`

Code:

`github.com/MukulRay1603/Empath-RAG`





Thank You

We hope you liked our presentation 😊

